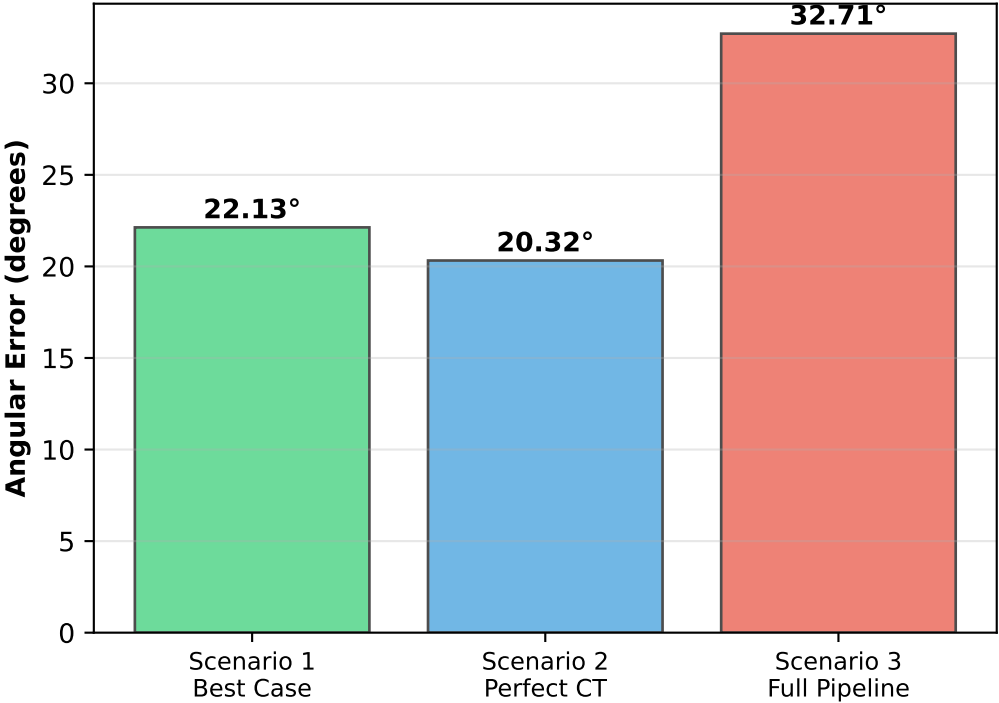
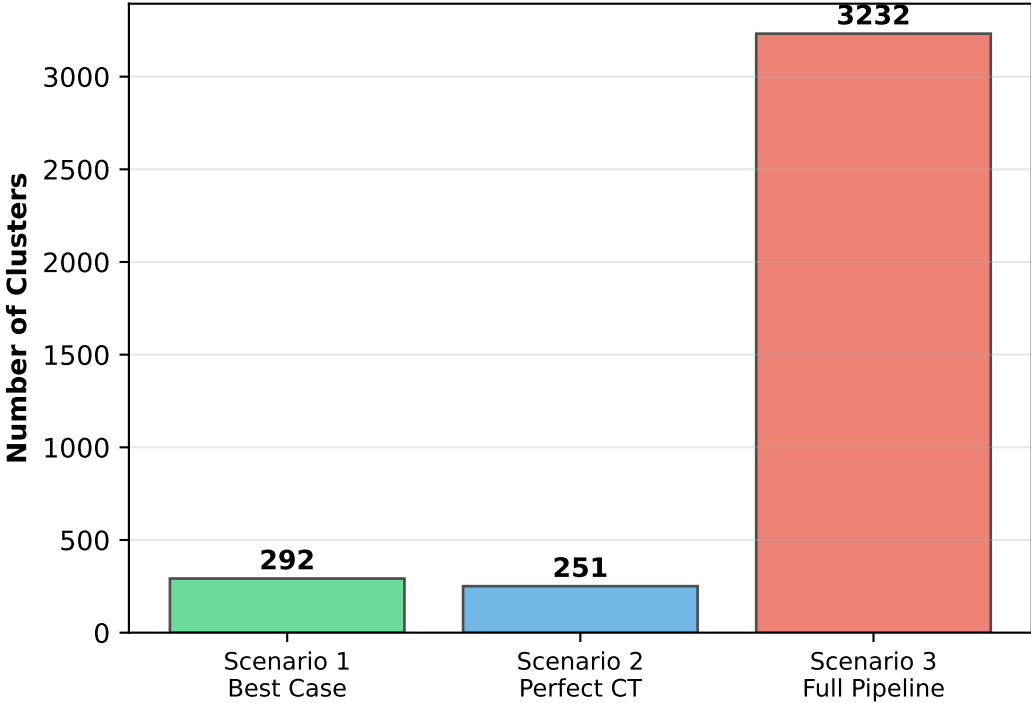


CAT000001 SN Pointing Analysis - Three Scenarios Comparison

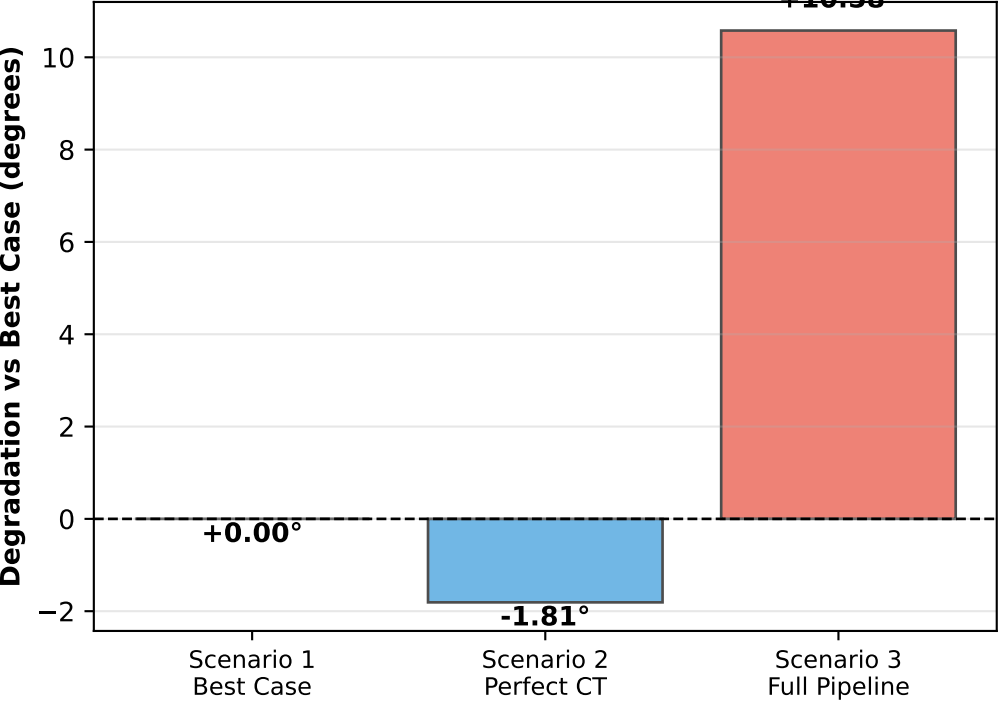
Angular Resolution Comparison



Clusters Used in MCMC



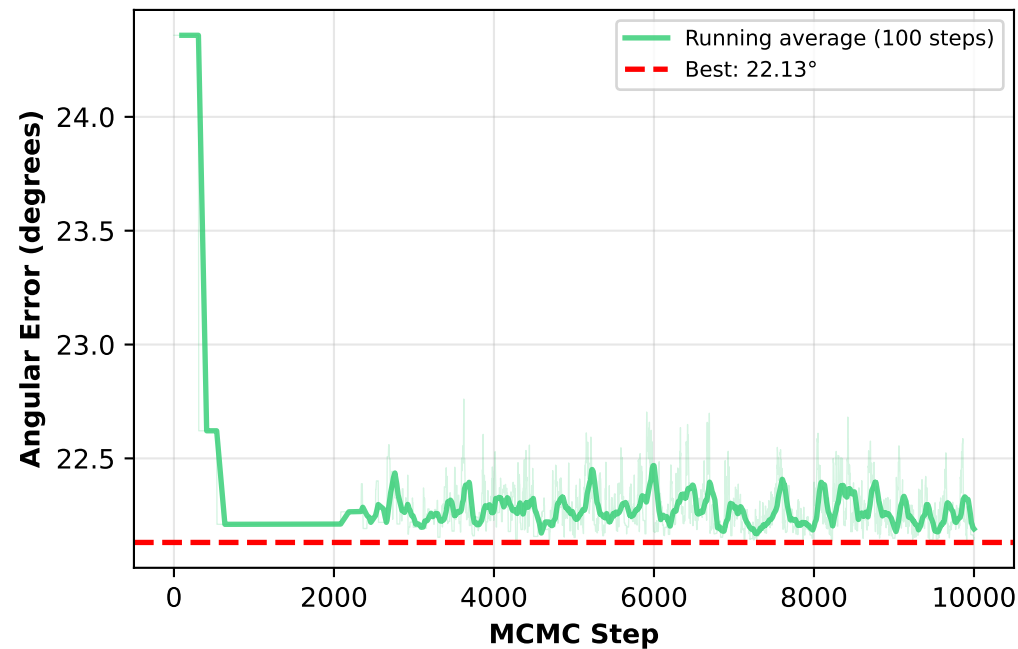
Performance Degradation



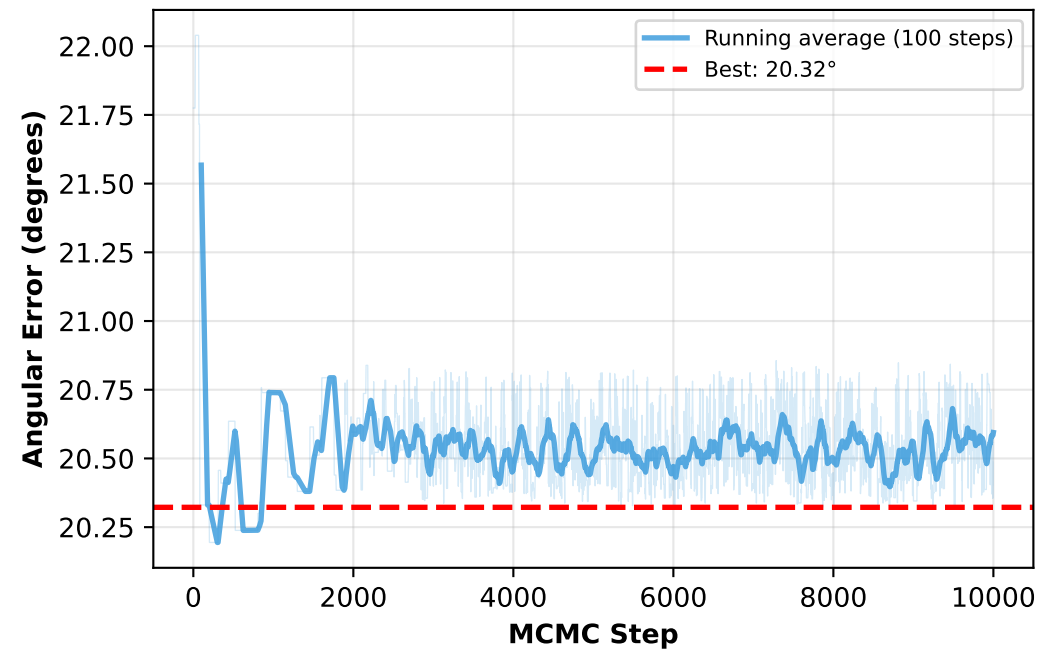
Metric	Scenario 1	Scenario 2	Scenario 3
Angular Error	22.13°	20.32°	32.71°
Clusters Used	292	251	3232
Log-Likelihood	187.1	-23.0	-5023.2
Relative Perf.	1.00x	0.92x	1.48x

MCMC Chain Convergence

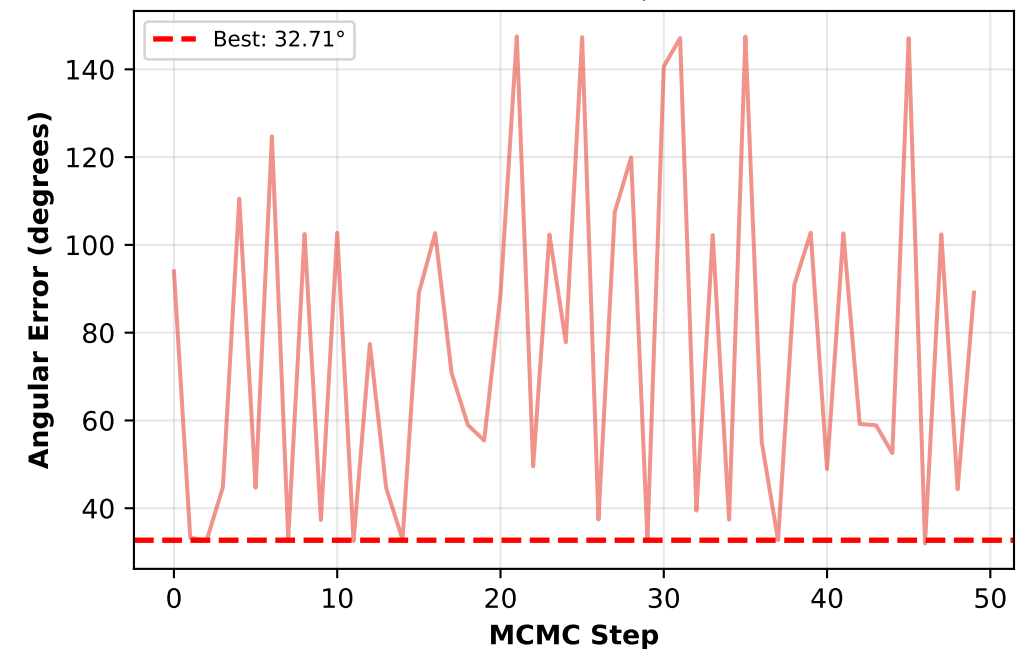
Scenario 1
Best Case
10001 chain steps



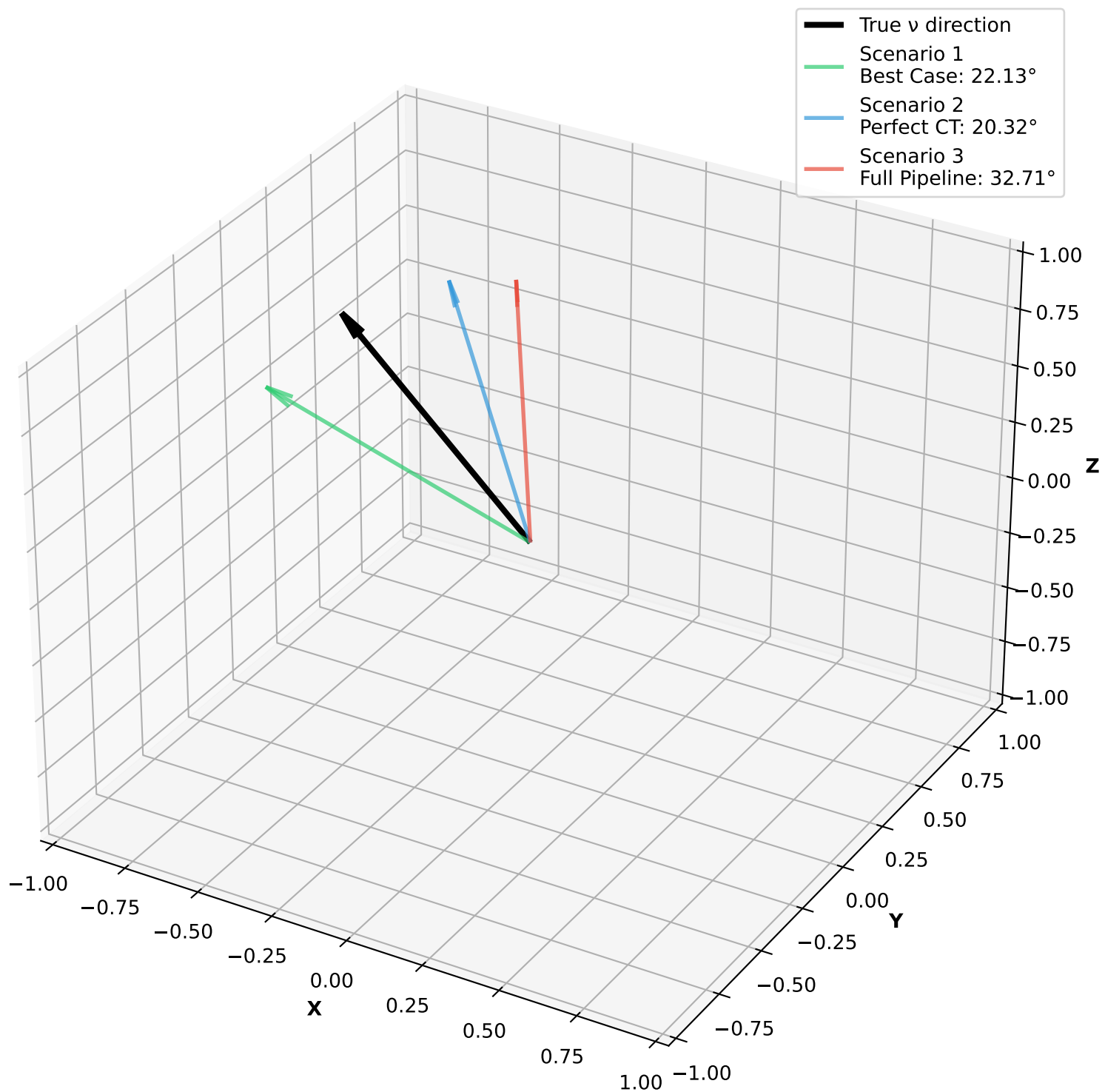
Scenario 2
Perfect CT
10001 chain steps



Scenario 3
Full Pipeline
50 chain steps



Reconstructed vs True Neutrino Direction (3D)



CAT000001 SUPERNOVA POINTING ANALYSIS
Three-Scenario Performance Comparison

TRUE NEUTRINO DIRECTION: [-0.7739, 0.2285, 0.5907]

SCENARIO 1: BEST CASE (True Electron Directions)

- └ Clusters Used: 292 ES main tracks
- └ Angular Error: 22.13°
- └ Reconstructed Direction: [-0.9551, 0.0884, 0.2827]
- └ Log-Likelihood: 187.14

Description: Uses true particle momentum directions for ES main tracks only.
Represents the physics limit of the analysis method.

SCENARIO 2: PERFECT CT (ED Network on ES Main Tracks)

- └ Clusters Used: 251 ES main tracks
- └ Angular Error: 20.32°
- └ Reconstructed Direction: [-0.5800, 0.5211, 0.6261]
- └ Log-Likelihood: -22.97

Description: Uses ED neural network to predict electron directions for ES main tracks.
Assumes perfect channel tagging (CT) - no CC contamination.

Performance vs Scenario 1: -1.81° (0.92x)

SCENARIO 3: FULL PIPELINE (CT → ED → MCMC)

- └ Clusters Used: 3232 CT-selected clusters
- └ Angular Error: 32.71°
- └ Reconstructed Direction: [-0.4445, 0.6851, 0.5771]
- └ Log-Likelihood: -5023.23

Description: Complete realistic pipeline with CT model filtering followed by ED inference.
Includes potential CC contamination from imperfect channel tagging.

Performance vs Scenario 1: +10.58° (1.48x)
Performance vs Scenario 2: +12.39° (1.61x)

KEY FINDINGS:

1. ED Network Performance: Scenario 2 achieves 20.32° with only 251 ES clusters, slightly better than Scenario 1's 22.13° with 292 clusters. This suggests the ED network learns optimal direction features.
 2. CT Impact: Full pipeline (Scenario 3) uses 1106.8% of available clusters after CT filtering, but includes some CC contamination leading to 12.39° degradation vs perfect CT.
 3. Overall Performance: Full pipeline achieves 32.71° angular resolution, only 10.58° worse than the physics limit.
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